

Turing Machine (Formal Definition)

A Turing Machine can be defined as a set of 7 tuples

$Q \rightarrow$ Non-empty set of states

$\Sigma \rightarrow$ Non-empty set of Symbols

$\Gamma \rightarrow$ Non-empty set of Tape Symbols

$\delta \rightarrow$ Transition function defined as

$$Q \times \Sigma \rightarrow \Gamma \times \{R/L\} \times Q$$

$q_0 \rightarrow$ Initial state

$b \rightarrow$ Blank symbol

$F \rightarrow$ Set of final states (Accept and Reject states)

Thus, the production rule of Turing M/C will be

$$\delta(q_0, a) \rightarrow (q_1, y, R)$$

Turing Thesis

Turing Thesis states that any computation that can be carried out by mechanical means can be performed by some Turing Machine.

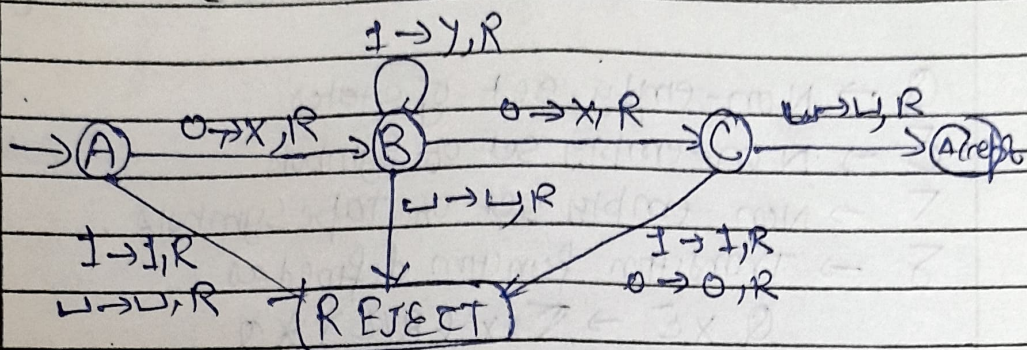
Recursively Enumerable Language

A language L and Σ^* is said to be Recursively

enumerable if there exists a Turing M/C that accepts it.

Q Design a Turing M/A which recognizes the language.

$$L = 0^i 1^j 0^i$$



0 1 1 0

x	y	y	x	blank
---	---	---	---	-------

Tape

Q $L = 0^N 1^N$

0	0	0	0	1	1	1	1	blank	blank
---	---	---	---	---	---	---	---	-------	-------

Algorithm

① Change "0" to "x"

② Move Right to first "1"

IF None: REJECT

③ Change "1" to "y"

④ Move left to leftmost "0"

⑤ Repeat the above steps until no more "0"s

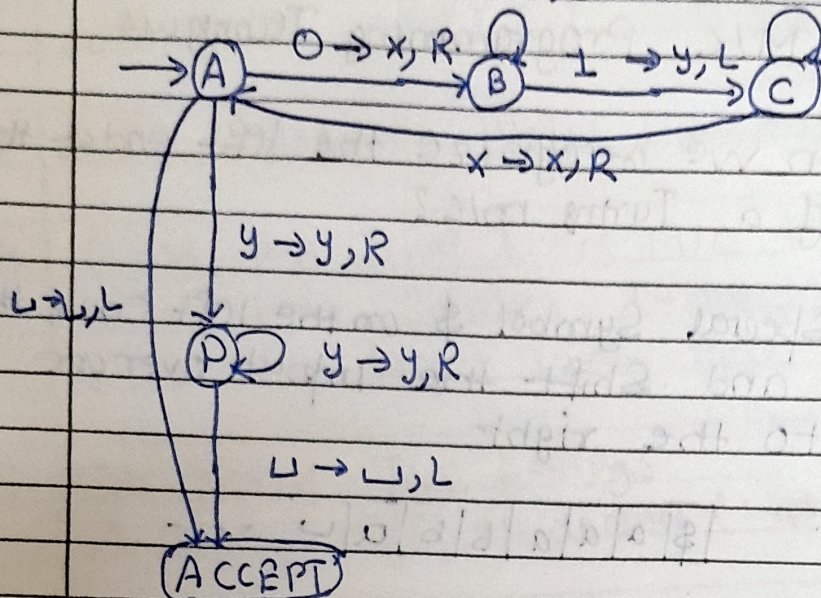
⑥ Make sure no more "1"s remain

Date ___/___/___

$y \rightarrow y, R$
 $0 \rightarrow 0, R$

$y \rightarrow y, L$
 $0 \rightarrow 0, L$

Page No. _____



The CHURCH - TURING Thesis.

Several variations of Turing m/c

- ① one tape or many
- ② Infinite on both ends
- ③ Alphabets only 2, 0, 1 or more.
- ④ Can the Head also stay in the same place
- ⑤ Allow Non-Determinism

All variations are
 equivalent in
 computing capability

Turing M/C and lambda calculus are equivalent
 in power.

Algorithmically computable
 Means

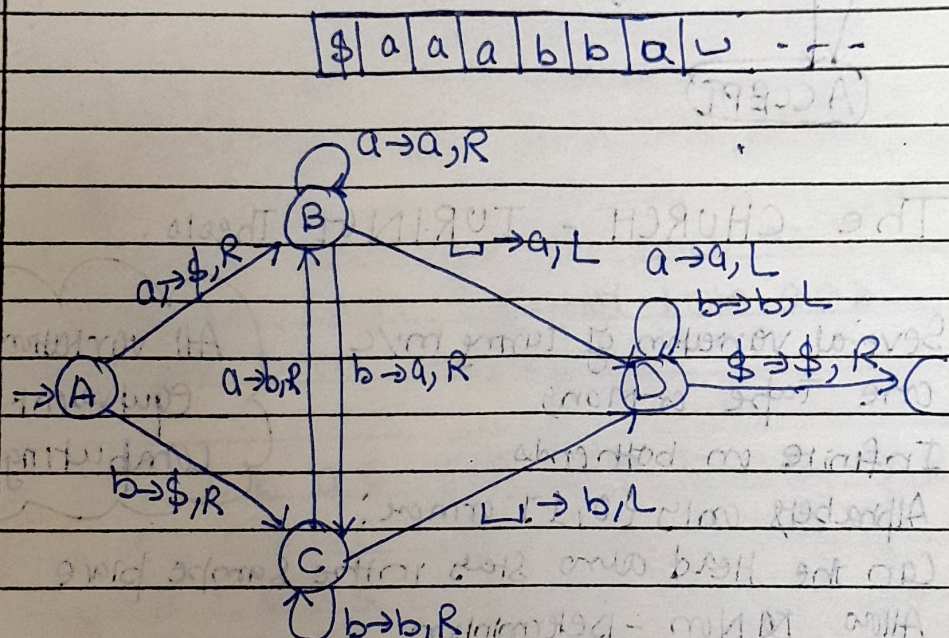
Computable by a TURING MACHINE

Turing M/C $\neq$ Turing Test

Turing M/C Programming Techniques

Q How can we recognize the left end of the Tape of a Turing m/c?

Put a Special Symbol \$ on the left end of the tape and Shift the input over one cell to the right



Q Build a Turing M/C to recognize the language $0^N 1^N 0^N$

Idea

We already have a Turing M/C to turn $0^N 1^N$ to $x^N y^N$ and to decide that language

Use this Turing M/C as a subroutine

Step 1

0 0 0 0 0	1 1 1 1 1	0 0 0 0 0
x x x x x	y y y y y	0 0 0 0 0

Step 2 Build a similar Turing M/C to recognize 'y' or 'n'

Step 3 Build the final Turing M/C by combining these two smaller Turing M/C together into one larger Turing M/C

Q Comparing Two strings
A Turing M/C to decide $\{w \# w \mid w \in \{a, b, c\}^*\}$

① use a new symbol such as 'x'

② Replace each symbol into an 'x' after it has been examined

a b b a c	#	a b b a c	} But we lost the original string.
x b b a c	#	x b b a c	
x x b a c	#	x x b a c	
x x x a c	#	x x x a c	
x x x x c	#	x x x x c	
x x x x x	#	x x x x x	

Can we do it non-destructively? i.e., without losing the original strings?

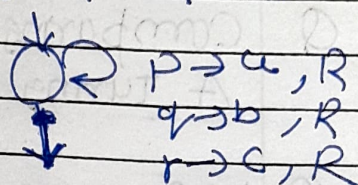
Replace each unique symbol with another unique symbol instead of replacing all with the same symbol.

Eg:- $a \rightarrow p$
 $b \rightarrow q$
 $c \rightarrow r$.

$a \ b \ b \ a \ c \ \# \ a \ b \ b \ a \ c$
 $\downarrow$

$p \ . \ q \ . \ q \ . \ p \ r \ \# \ p \ . \ q \ . \ q \ . \ p \ r$

Restore the original strings if required



Multitape Turing M/C

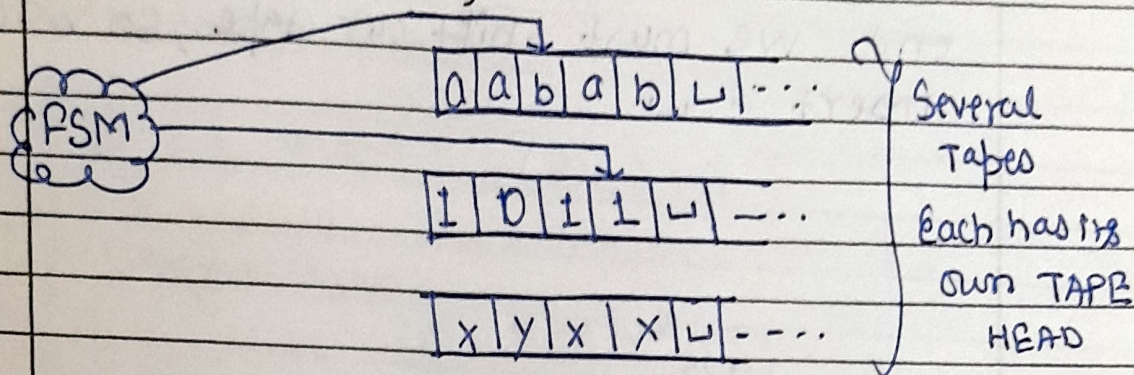
Theorem:- Every Multitape Turing M/C has an equivalent single tape Turing M/C

Proof.

Given a Multitape Turing M/C show how to build a single tape Turing M/C

- Need to store all tapes on a single tape
Show data representation
- Each tape has a tape head
Show how to store that info
- Need to Transform a move in the Multitape TM into one or ~~more~~ more moves in the Single Tape TM

Multitape Turing M/C



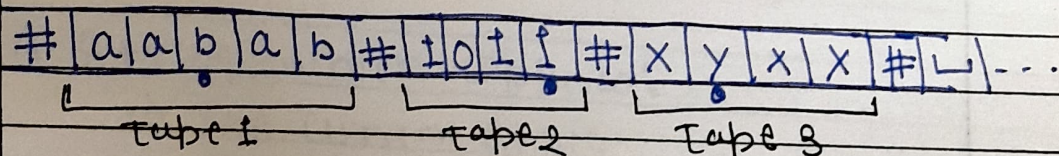
Q $\xrightarrow{b1y \rightarrow a0x, L, R}$ R



An Example M/C with
 $K=3$ Tapes.

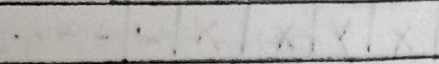
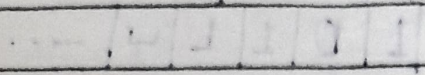
Single Tape Turing M/C

FSM $\leftarrow$ A TM with one Tape
to simulate the above Multitape
TM



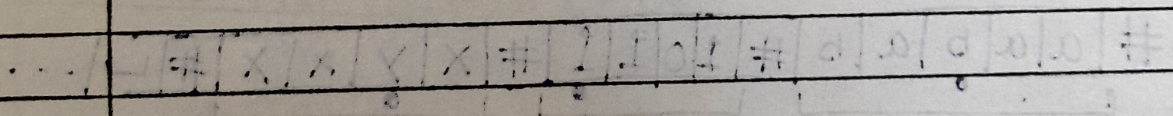
- Add "dots" to show where Head "k" is
- To simulate a transition from state Q, we must scan our Tape to see which Symbols are under the K Tape Heads
- Once we determine this and are ready to MAKE the Transition, we must scan across the tape again to update the cells and move the dots.

Whenever one head moves off the right end, we must shift our tape, so we can insert a $\perp$



A TM with one tape

to determine the above results



• Add "Gib" to show where "Hob" is.

To eliminate a horizontal band effect, we must feed our tape to see which symbols are under the A tape head.

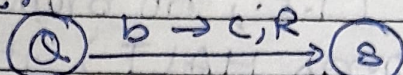
[illegible]

Non Determinism in Turing M/C

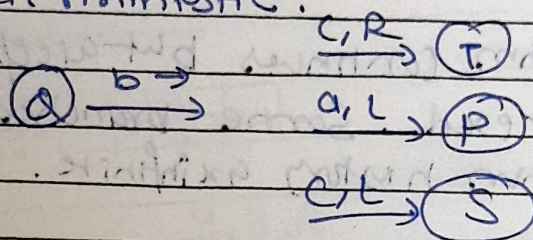
Transition function..

$$Q \times \Sigma \Rightarrow P \{ \tau \times (R/L) \times Q \}$$

Deterministic:



Non-Deterministic:



Configuration

- A way to represent the entire state of a TM at a moment during computation.
- A string, which captures:
 - The current state
 - The current position of the head
 - The entire tape contents.

a b a b b a a L

Q b a Q b b a Q

outcome of a Non deterministic Computation.

Accept If any Branch of the Computation accepts then the non deterministic TM will accept

REJECT If all branches of the Computation HALT and REJECT (i.e., no branches accept, but all Computations HALT) then the Nondeterministic TM Rejects

LWP Computation continues but accept is never encountered. Some branches in the computation history go infinite.

Turing M/C as Problem Solver.

Any arbitrary Problem can be expressed as a language

- Any Instance of the problem is encoded into a String

The String is in the language $\rightarrow$ The ans is yes

The String is ^{not} in the lang. $\rightarrow$ The ans is NO

$$L = a^n b^n$$

possible String

$$L = \{ ab, aabb, aaabbb \dots \}$$

→ Let take $n=4$ and understand the working step by step.

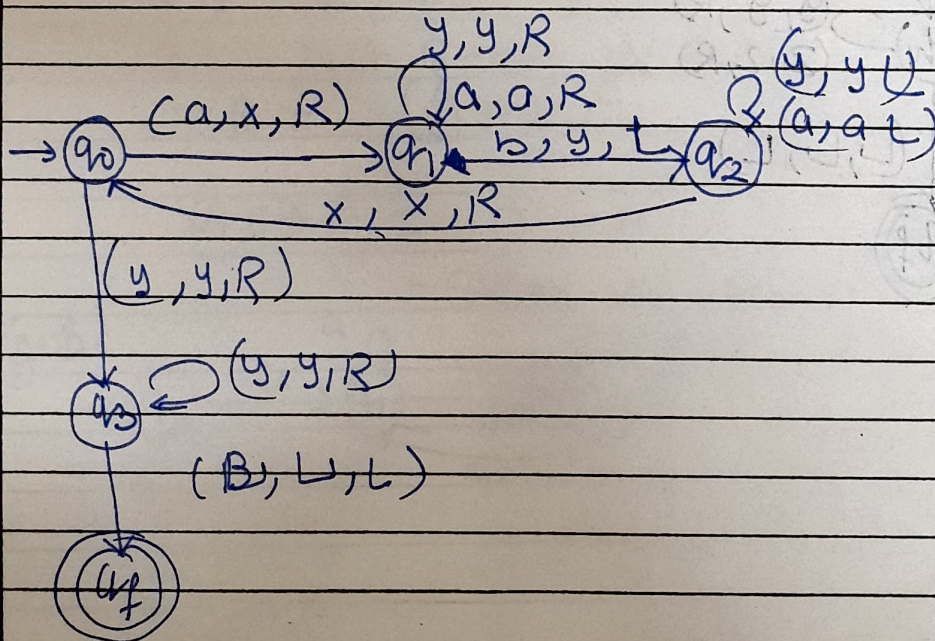
$$\text{for } n=4 \Rightarrow aaabbbb$$

$$x \cancel{a} a a \cancel{b} b b b \rightarrow x \cancel{a} a a y \cancel{b} b b$$

$$x x x \cancel{a} y y y \cancel{b} y \leftarrow x x \cancel{a} y y \cancel{b} b$$

$$\Rightarrow x x x x y y y y$$

$$_ _ _ _ | a | a | a | a | b | b | b | b | _ _ _ _$$

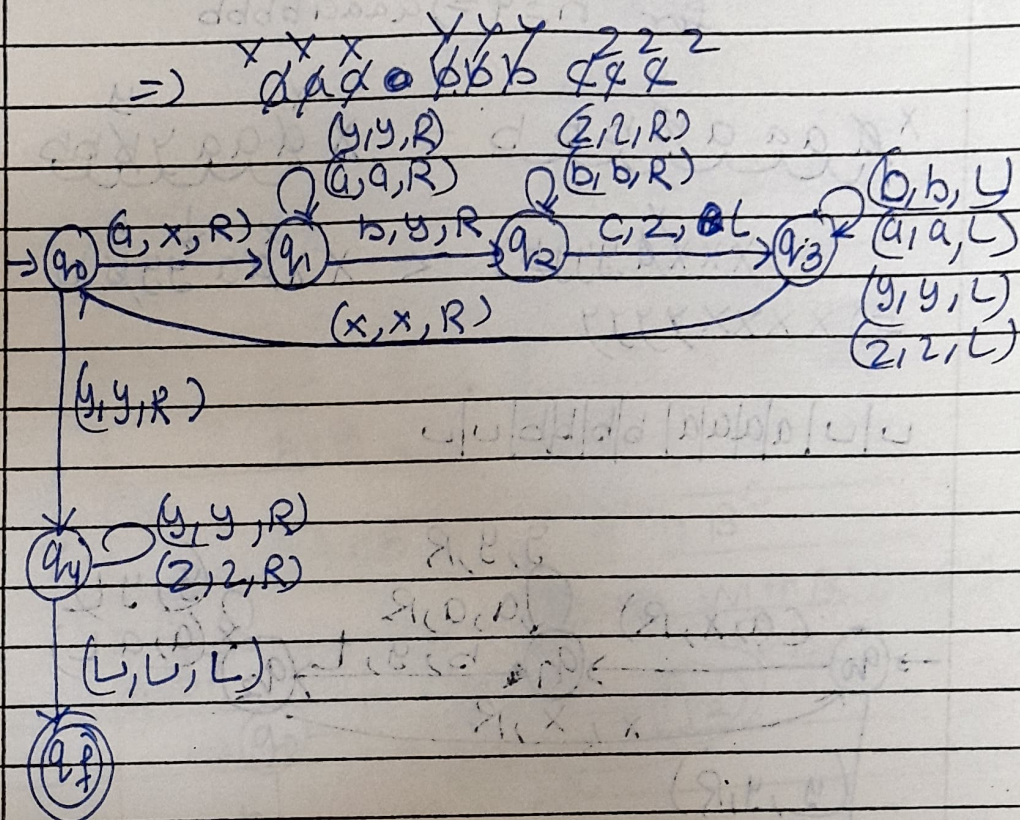


Q $L = a^n b^n c^n$

$L = abc, aabbcc, aaabbbccc,$

$\rightarrow$ let's take $n=3$

$\Rightarrow$ i.e., $aaabbbccc$



$acabc$

$$a(b+ca)^*c$$

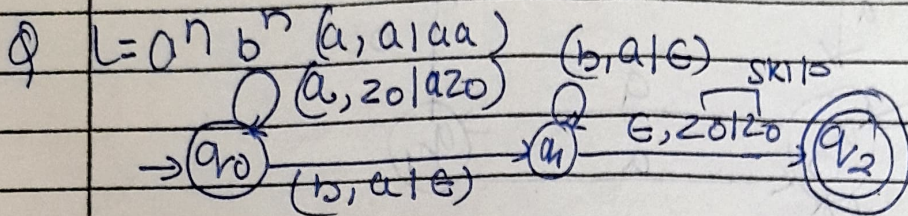
$$a b^* c$$

$$a(b+ca)^*c$$

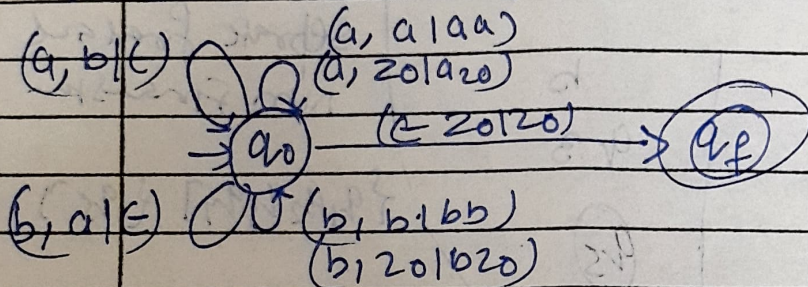
Date / /

Page No.:

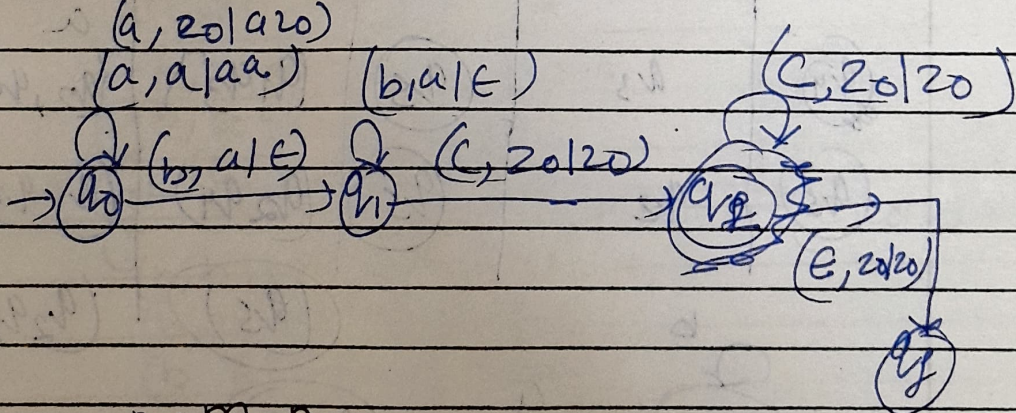
Pd a.



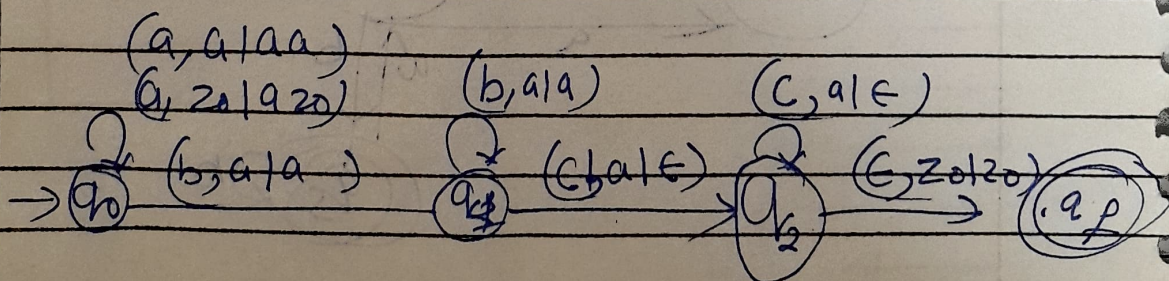
Q $L = w \mid n_a(w) = n_b(w)$



Q $L = a^n b^n c^n$



Q $L = a^n b^m c^n$

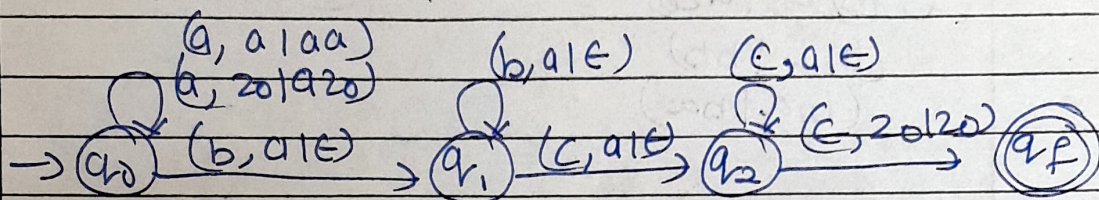


$$L = a^{m+n} b^m c^n$$

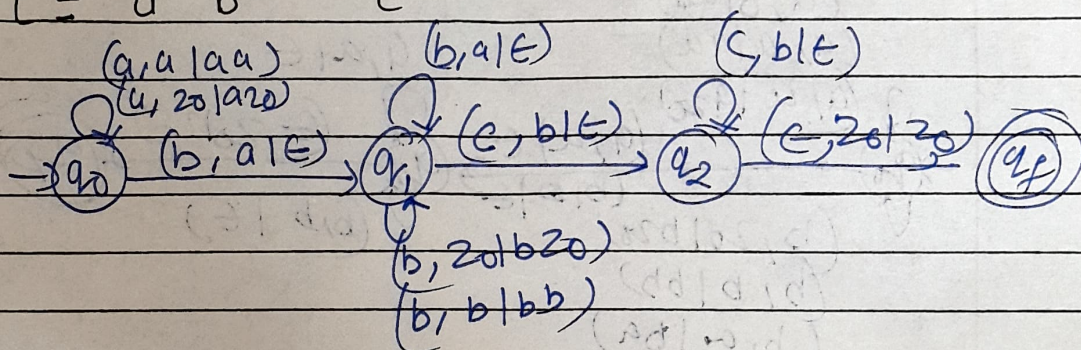
$$n=1 \quad m=1 \quad L = a a b c$$

$$n=1 \quad m=2 \quad L = a a a b b c$$

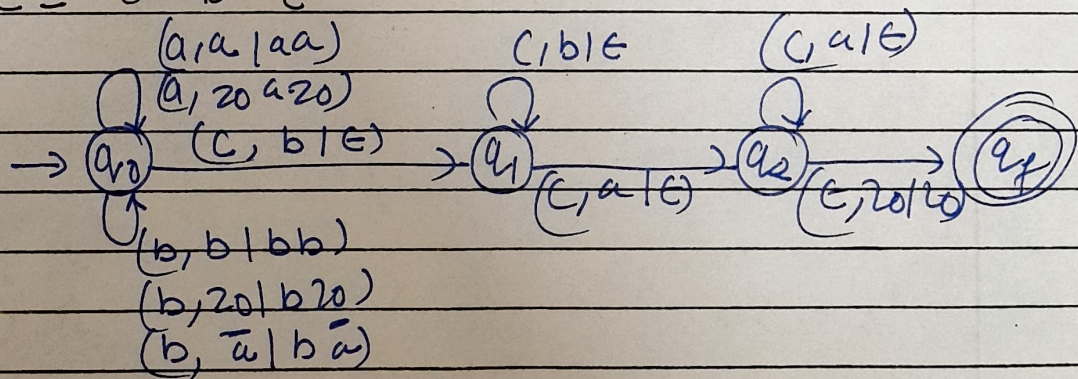
$$n=2 \quad m=1 \quad L = a a a b c c$$



$$L = a^n b^{m+n} c^m$$



$$L = a^n b^m c^{m+n}$$

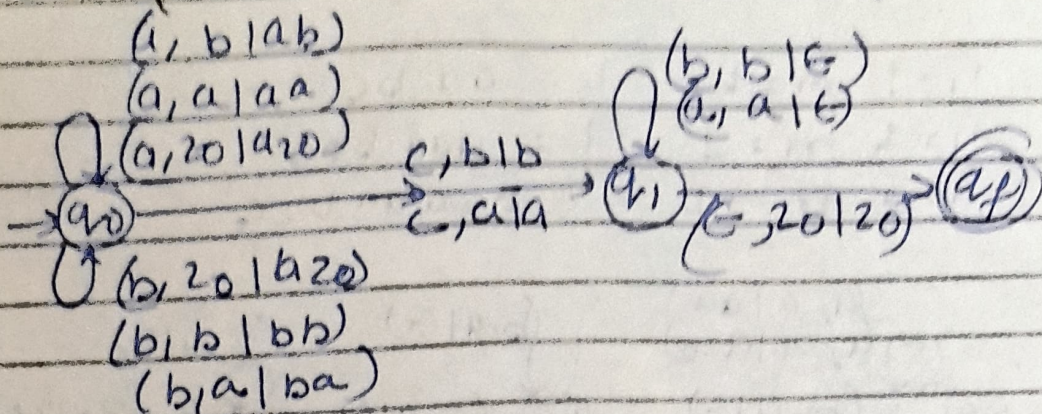


Imp

$$w = abb \quad wR = bba \quad w \cup R = abb \cup bba$$

Page No.

$$L = \{ w \cup R \mid w \in \{a, b\}^* \}$$



Imp

$$L = \{ w \cup R \mid w \in \{a, b\}^* \}$$

